

Full Sharded Data Parallel

Lorem Ipsum

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11. Sharding: split one big thing into pieces

One large model state

MODEL STATE

Story intuition

In games, shards are small pieces that together complete one item. FSDP uses the same idea for training state.

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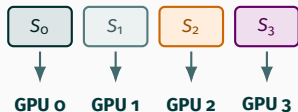
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- Give different pieces to different GPUs.

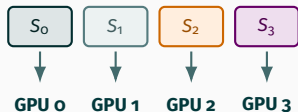
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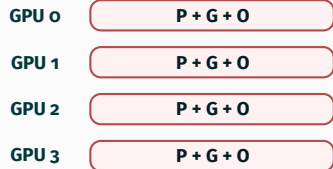
In games, shards are small pieces that together complete one item. FSDP uses the same idea for training state.

- Split one large state into pieces.
- Give different pieces to different GPUs.
- Reconstruct a full piece of the model only when computation needs it.

FSDP starts by **splitting** model state instead of **replicating** it.

12. What exactly is sharded?

DDP: replicated



Every GPU keeps a full copy.

FSDP: sharded

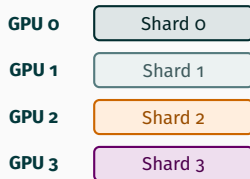
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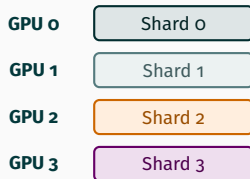
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P = parameters (weights), **G = gradients**, **O = optimizer states** (for Adam, moment estimates). FSDP can shard all three persistent model states.

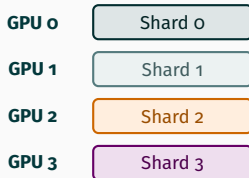
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Idealized persistent model-state memory per GPU scales roughly as $\frac{P+G+O}{N}$.

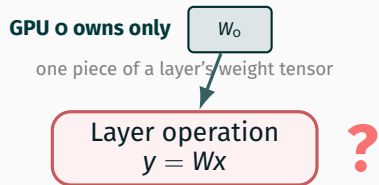
13. The first problem: a layer cannot compute from one shard

GPU 0 owns only

W_0

one piece of a layer's weight tensor

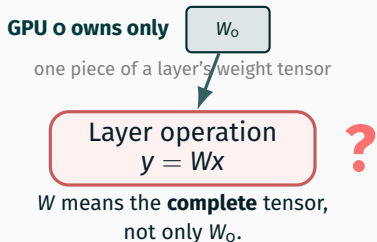
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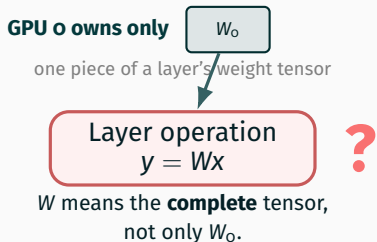
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$$W_0 + W_1 + W_2 + W_3$$

```
graph LR; W0[W0] --- P1[+]; P1 --- W1[W1]; W1 --- P2[+]; P2 --- W2[W2]; W2 --- P3[+]; P3 --- W3[W3];
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$$W_0 + W_1 + W_2 + W_3$$

So FSDP must **temporarily collect** the missing parameter shards before computation.

14. All-Gather: temporarily rebuild the current FSDP unit

Before: each GPU owns one shard

GPU 0



GPU 1



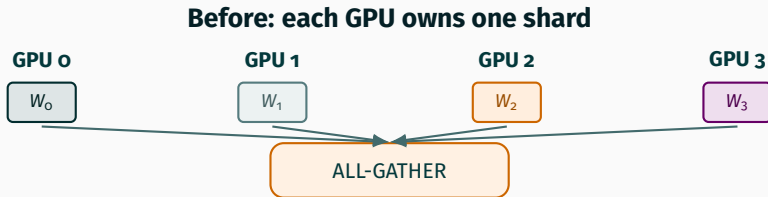
GPU 2



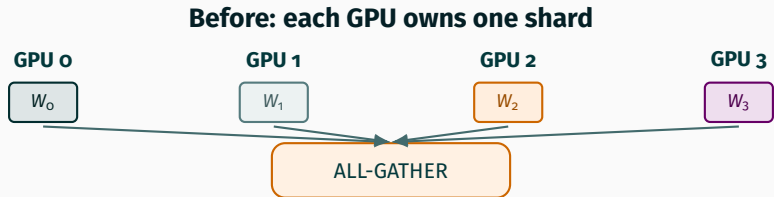
GPU 3



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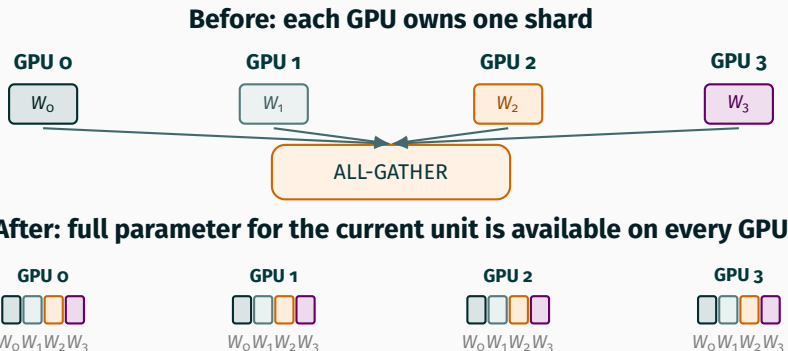
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After: full parameter for the current unit is available on every GPU

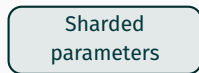


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All-Gather: **shards** → **full parameters**, but only for the unit currently being computed.

15. Forward pass: gather, compute, reshard



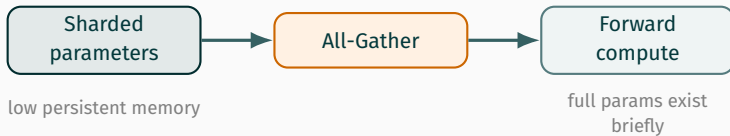
low persistent memory

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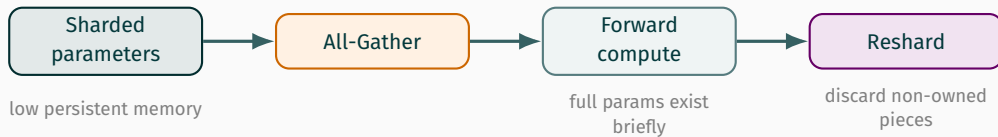


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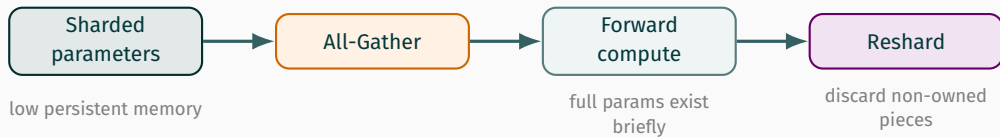
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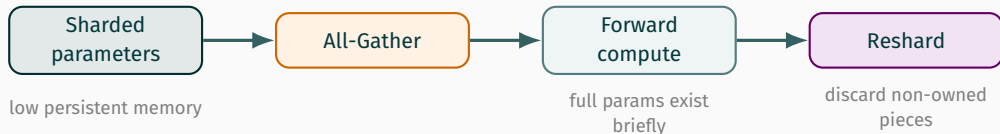
Resharding

After the forward computation, each GPU frees the full parameter and keeps only the shard it owns.

Why it saves memory

The expensive full representation is temporary; the sharded representation is the persistent one.

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Resharding

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Forward = **All-Gather** → **compute** → **reshard**.

16. Why gathering does not destroy the memory benefit

Layer-wise FSDP units

Transformer block 1

Transformer block 2

Transformer block 3

Transformer block 4

Other blocks can stay sharded.

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Block 1: full

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Important distinction

FSDP gathers only the **current FSDP unit** (for example, one Transformer block), not the entire model.

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Peak memory is not simply $(P + G + O)/N$. Activations and temporary full-parameter buffers also occupy memory.

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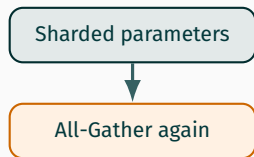
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FSDP controls peak memory by gathering **one unit at a time**.

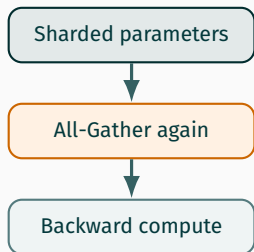
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Sharded parameters

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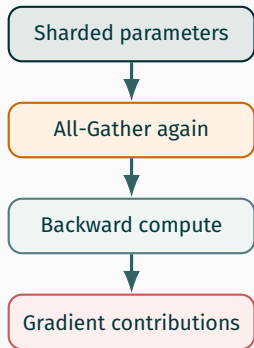
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Why gather again?

Backpropagation needs the current layer's parameters to compute gradients with respect to inputs and weights.

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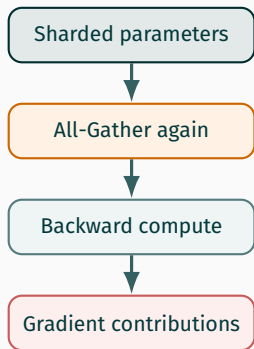
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Backward creates distributed gradient contributions; FSDP must combine them efficiently.

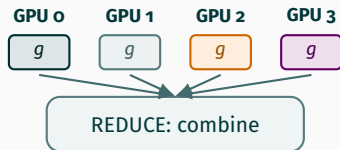
18. Reduce-Scatter: combine gradients, then split again

1. Gradient contributions



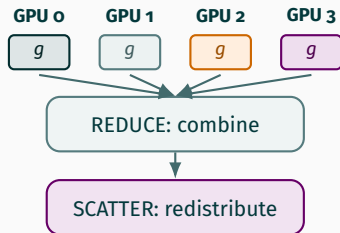
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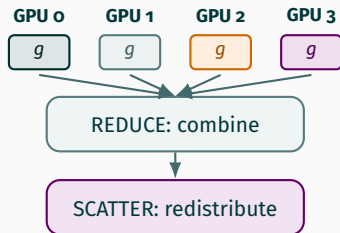
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Correct global gradient,
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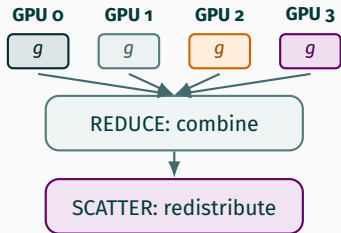
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Compare with DDP

DDP typically uses **All-Reduce**: every rank ends with the full synchronized gradient. FSDP uses **Reduce-Scatter**: each rank ends with only its shard.

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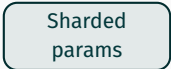
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Reduce-Scatter = **combine** gradient contributions, then **redistribute** only the owned shard.

19. The complete FSDP training cycle



Sharded
params

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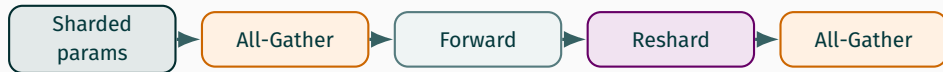
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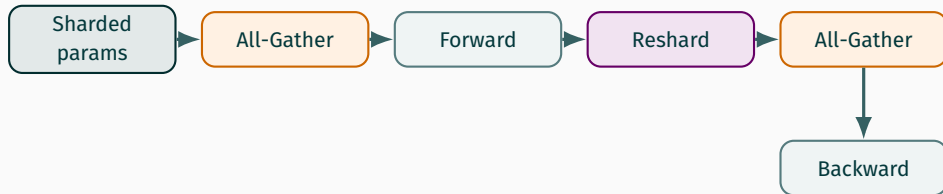
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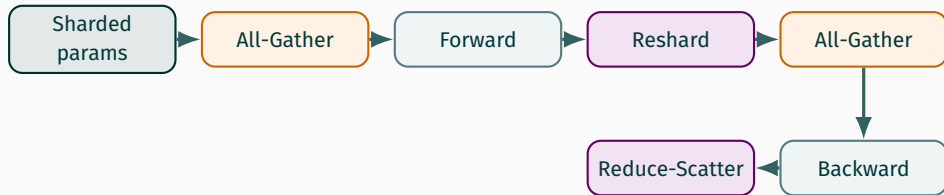
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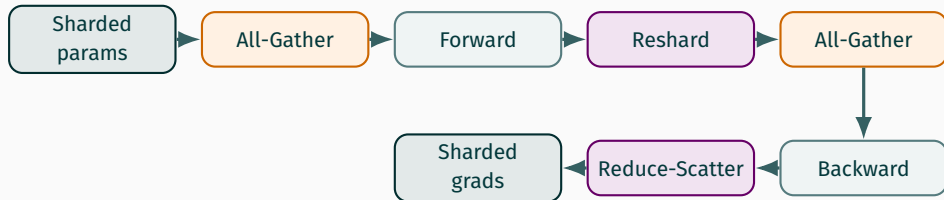
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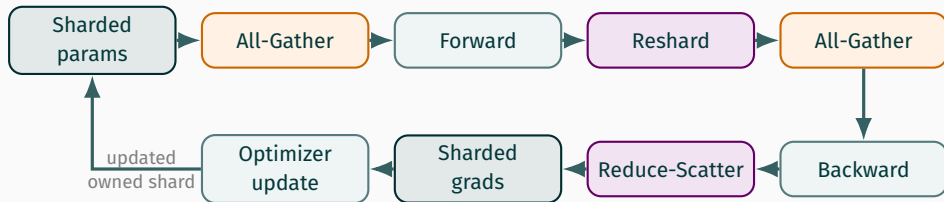
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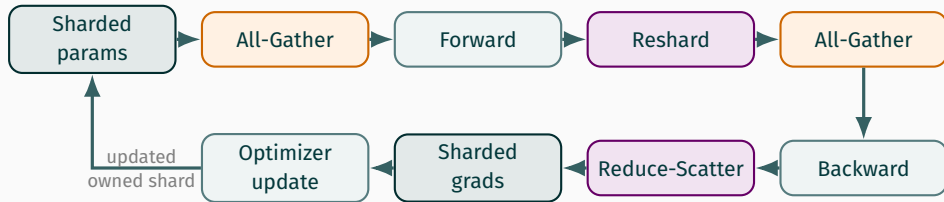
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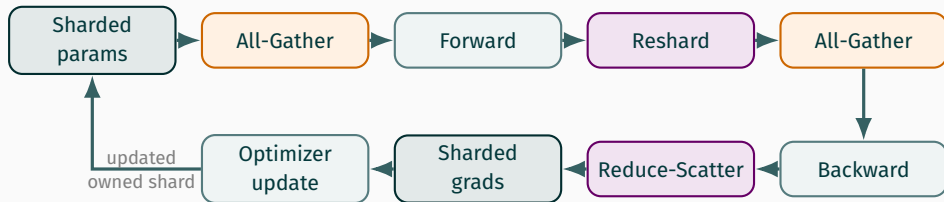


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Keep model state divided for memory; gather parameters briefly only when computation needs them.

20. Final trade-off: memory down, communication up

	DDP	FSDP
Model state	Replicated	Sharded
Memory/GPU	Higher	Lower
Parameters	Already local	All-Gather
Gradient sync	All-Reduce	Reduce-Scatter
Main pressure	Memory	Communication

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Compute block i

Gather block $i + 1$

Communication for the next unit can overlap with current computation.

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FSDP enables larger-model training by trading **memory** for **communication**.